

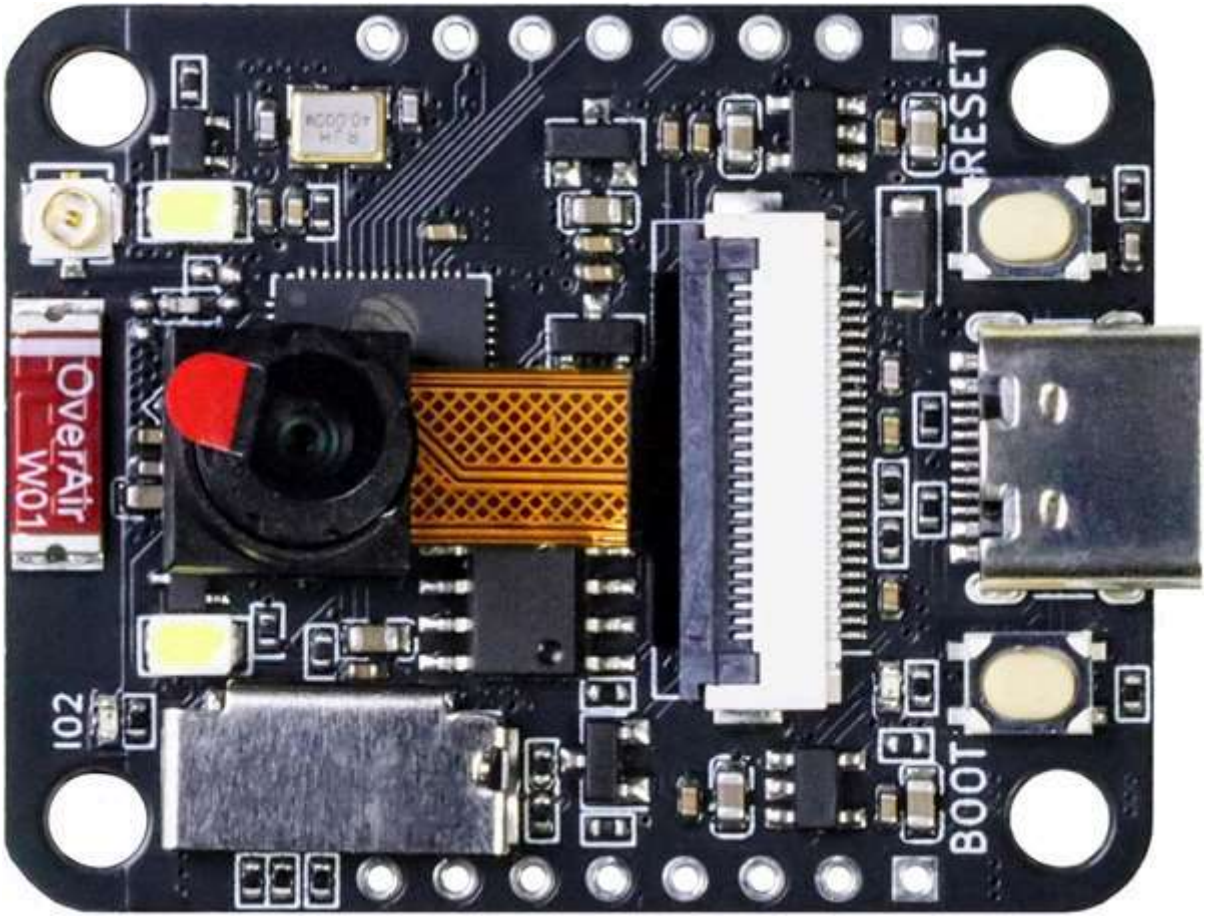
**NL-leishen** add flash light pin eb06132 · 4 months ago **5 Commits**

| | | |
|---|---|---|
| picture | update document | 7 months ago |
| README.md | add flash light pin | 4 months ago |
| esp32s3_cam_sch.pdf | update document | 7 months ago |

README

# ESP32-S3-CAM Development Board

# Introduction




The **ESP32S3-CAM** is a high-performance, low-power development board based on the Espressif **ESP32-S3** SoC. It features an integrated camera interface, TF card slot, and USB-C USB for easy debugging. Equipped with dual-core LX7 processors and AI acceleration capabilities, it is ideal for AIoT applications such as image recognition, wireless monitoring, and smart home gateways.

## Key Features

- **Powerful Processing:** Xtensa® 32-bit LX7 dual-core processor, up to 240MHz.

- **AI Acceleration:** Built-in vector instructions to accelerate neural network computing and signal processing.
- **Easy Programming:** Onboard USB-to-Serial (Native USB) via Type-C interface—no external downloader required.
- **Rich Storage:** Large 8MB PSRAM and 8MB Flash for complex image processing tasks.
- **Expandability:** Supports TF card (up to 16GMB) and features multiple GPIOs for sensors.
- **Seamlessly Compatibility:** Fully compatible with existing ESP32-S3-CAM models, providing access to a wealth of online tutorials and community resources.
- **Applications:** Home smart devices & video doorbells; Industrial wireless monitoring; QR code recognition & Face detection; DIY educational robot vision, and more.
- **Download:** Programming can be done directly via the USB-C port without requiring an external download module. Its usage is fully compatible with commercially available ESP32S3-CAM models.

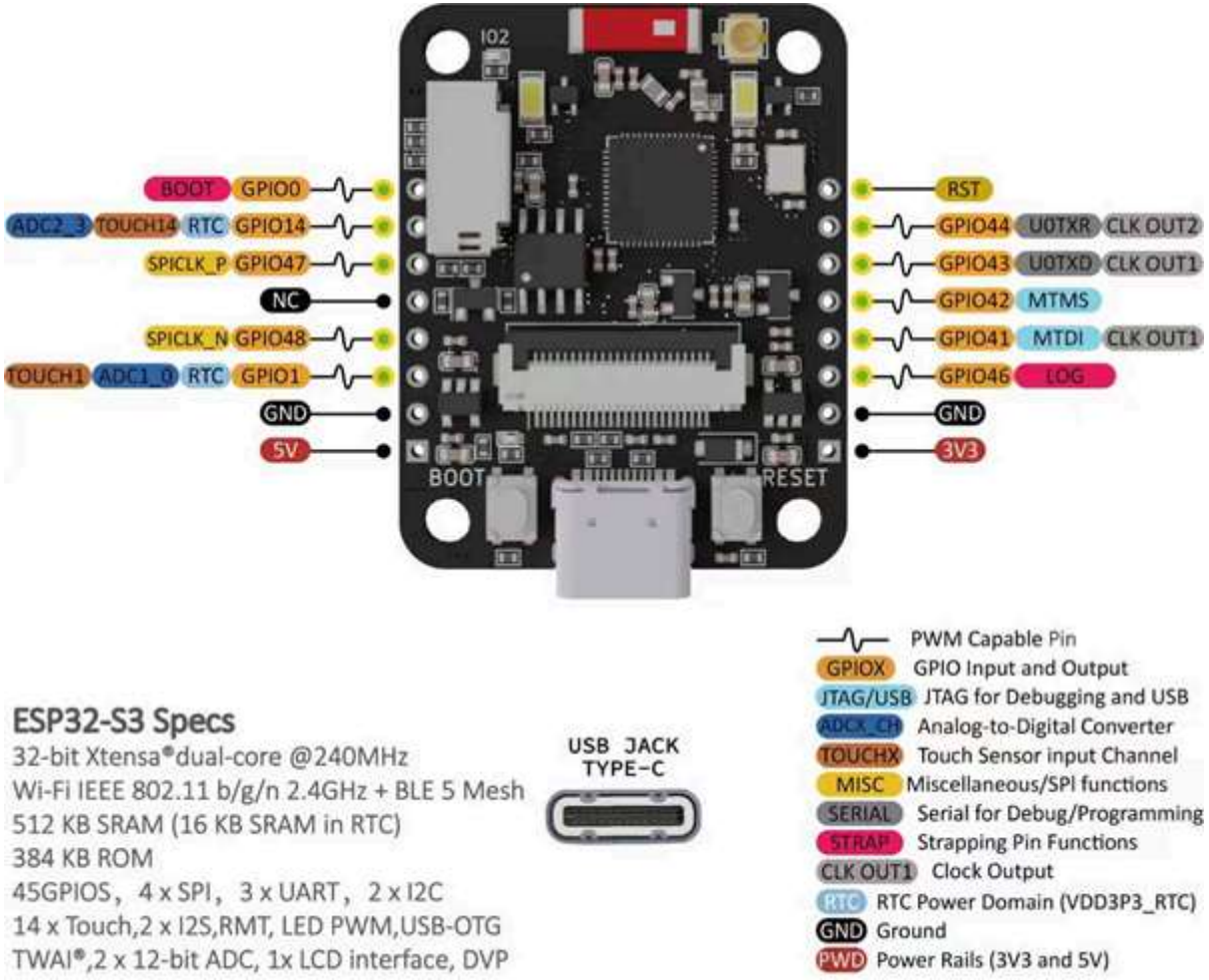
# Hardware

## Specifications

| Feature | Specification |
|---|---|
| **Microcontroller** | ESP32-S3R8 (Dual-core, 240MHz) |
| **Wireless** | 2.4GHz Wi-Fi & Bluetooth 5 (LE) |
| **SRAM/PSRAM** | 512KB SRAM / 8MB Octal PSRAM |
| **Flash** | 8MB |
| **Support Camera** | OV3660/0V2640 |
| **Interface** | Type-C USB, TF Card Slot, PH2.0 Battery JST |
| **Onboard Peripherals** | Reset/Boot Buttons, Double LED Flashlight, Power Indicator, Charging Circuit |

| Feature | Specification |
| --- | --- |
| **Digital I/O** | GPIO, SPI, I2C, UART, ADC, PWM |
| **Flash light pin** | GPIO3 |

## Pin Definition




- **Camera Interface:** DVP 8-bit interface.

- **TF Card:** SPI/SDIO (Check schematic for specific GPIO mapping).
- **USB:** GPIO 19 (D-), GPIO 20 (D+).

### Camera pin

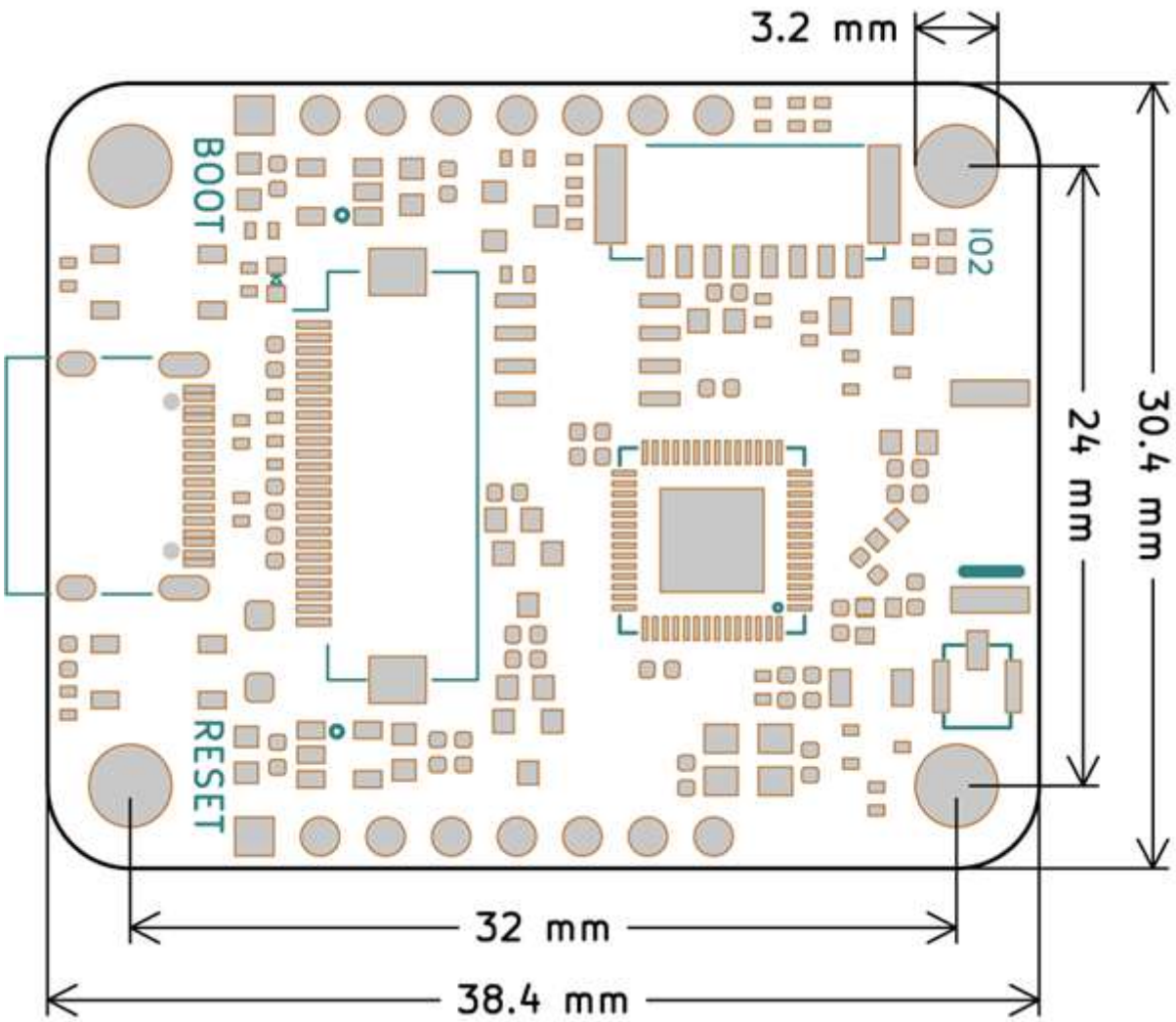
| CAMERA | ESP32S3 |
|---|---|
| I2C_SDA | GPIO4 |
| I2C_SCL | GPIO5 |
| CAM_RST | |
| DVP_VSYNC | GPIO6 |
| DVP_HREF | GPIO7 |
| DVP_Y9 | GPIO16(XTAL_32K_N) |
| XMCLK | CPIO15(XTAL_32K_P) |
| DVP_Y8 | GPIO17 |
| DVP_Y7 | GPIO18 |
| DVP_PCLK | GPIO13 |
| DVP_Y6 | GPIO12 |
| DVP_Y5 | GPIO10 |
| DVP_Y4 | GPIO8 |
| DVP_Y3 | GPIO9 |
| DVP_Y2 | GPIO11 |

### MMC Mode Pin

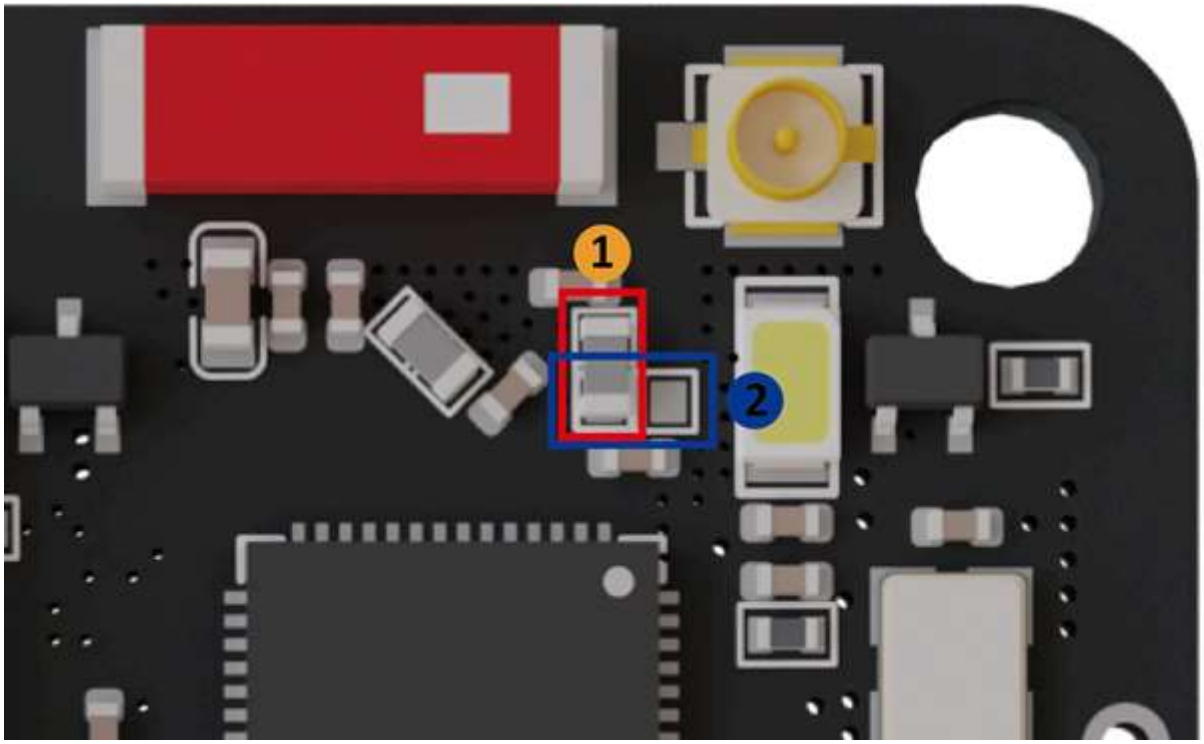
| MMC Mode | ESP32S3 |
| --- | --- |
| CMD | GPIO38 |
| CLK | GPIO39(MTCK) |
| DAT0 | GPIO40(MTD0) |
| DAT1 | NC |
| DAT2 | NC |
| DAT3 | NC |
| VDD | 3.3V |
| VSS | GND |

## Size

Download 3D file




## Antenna Switching Instruction

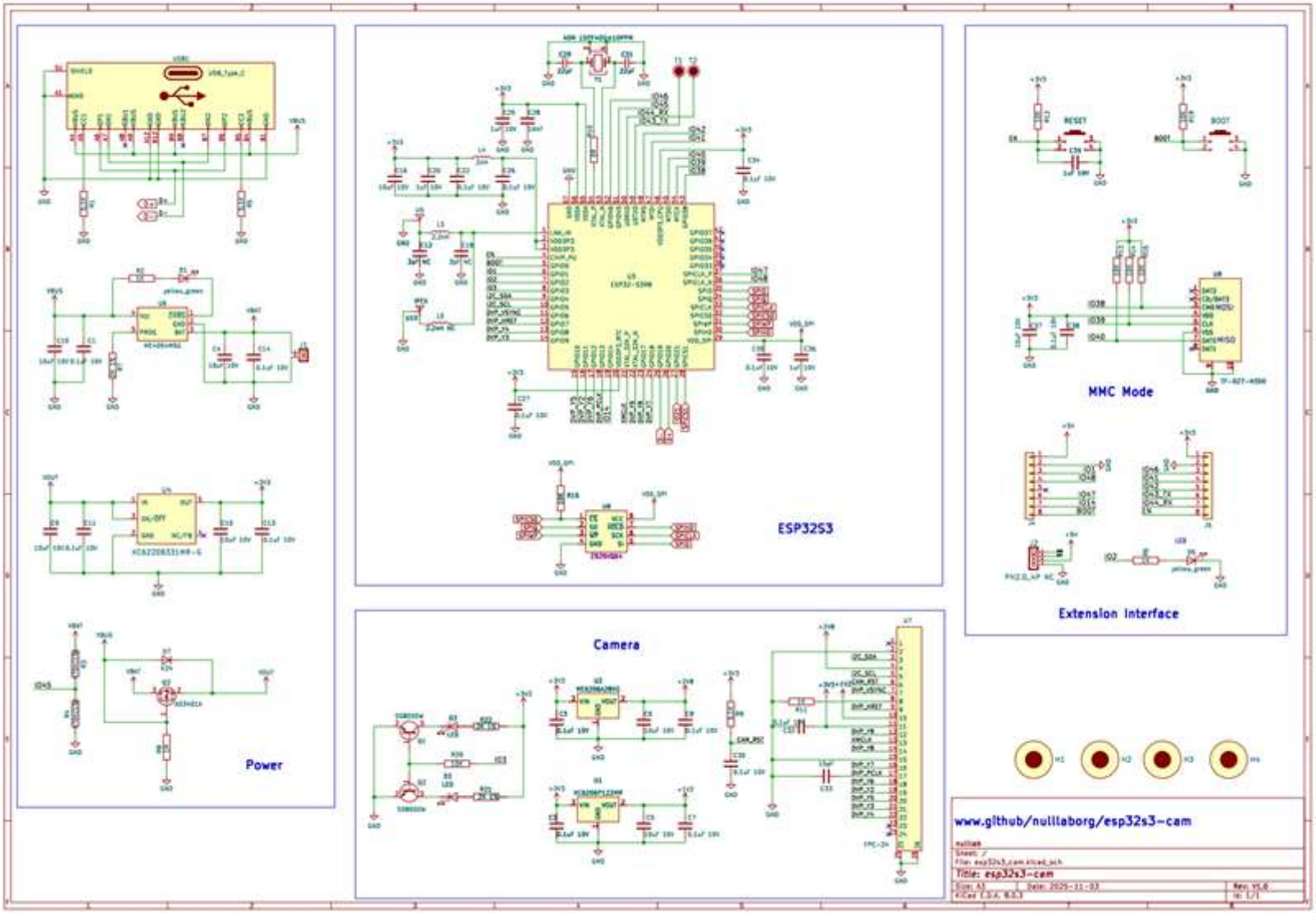



- **Option 1:** Use Onboard Antenna.
- **Option 2:** Use IPEX Antenna.

The ESP32-S3-CAM supports both the **Onboard PCB Antenna** and an **External IPEX Antenna**. By default, the board is configured to use the onboard antenna. To switch to an external antenna, you need to relocate the **0Ω resistor** near the IPEX connector.

## Schematics

Download the schematic file




# Examples

## Software Support

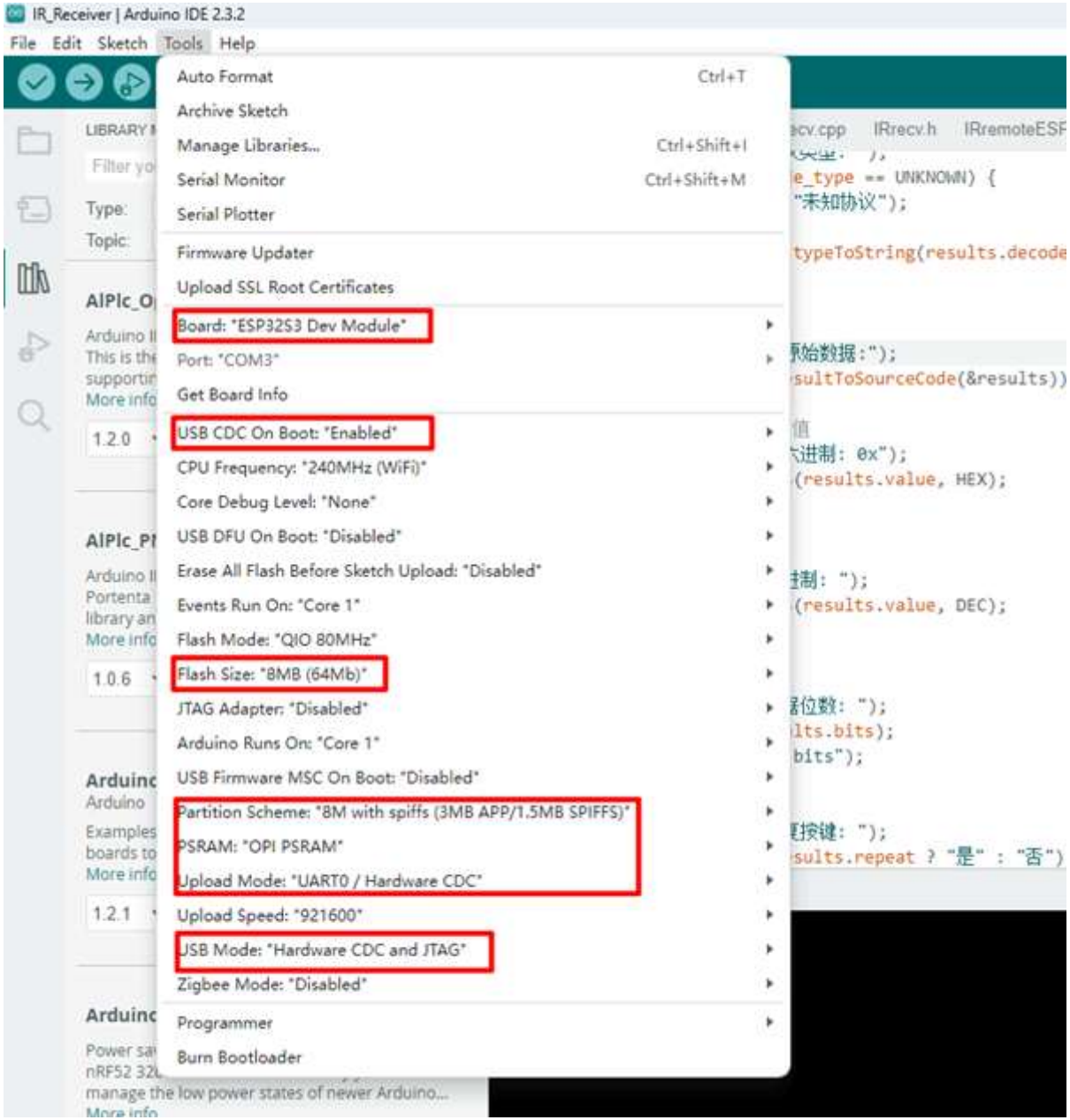
- **Official SDK:** ESP-IDF

- **Third-party Platforms:** Arduino IDE, MicroPython

## Configure In Arduino IDE

## Upload Examples: CameraWebServer

Arduino IDE Config: Board: Tools-->Board-->ESP32-->ESP32S3 Dev Module USB CDC On Boot: Tools-->USB CDC On Boot-->Enabled Flash Size: Tools-->flash Sizes-->8MHZ(64Mb) Partition Scheme: Tools-->Partition Scheme-->8M with spiffs(3MB APP/1.5MB SPIFFS) PSRAM: Tools-->PSRAM-->OPI PSRAM Upload Mode: Tools-->Upload Mode-->UART0/Hardware CDC USB Mode: Tools-->USB Mode-->Hardware CDC and JTAG

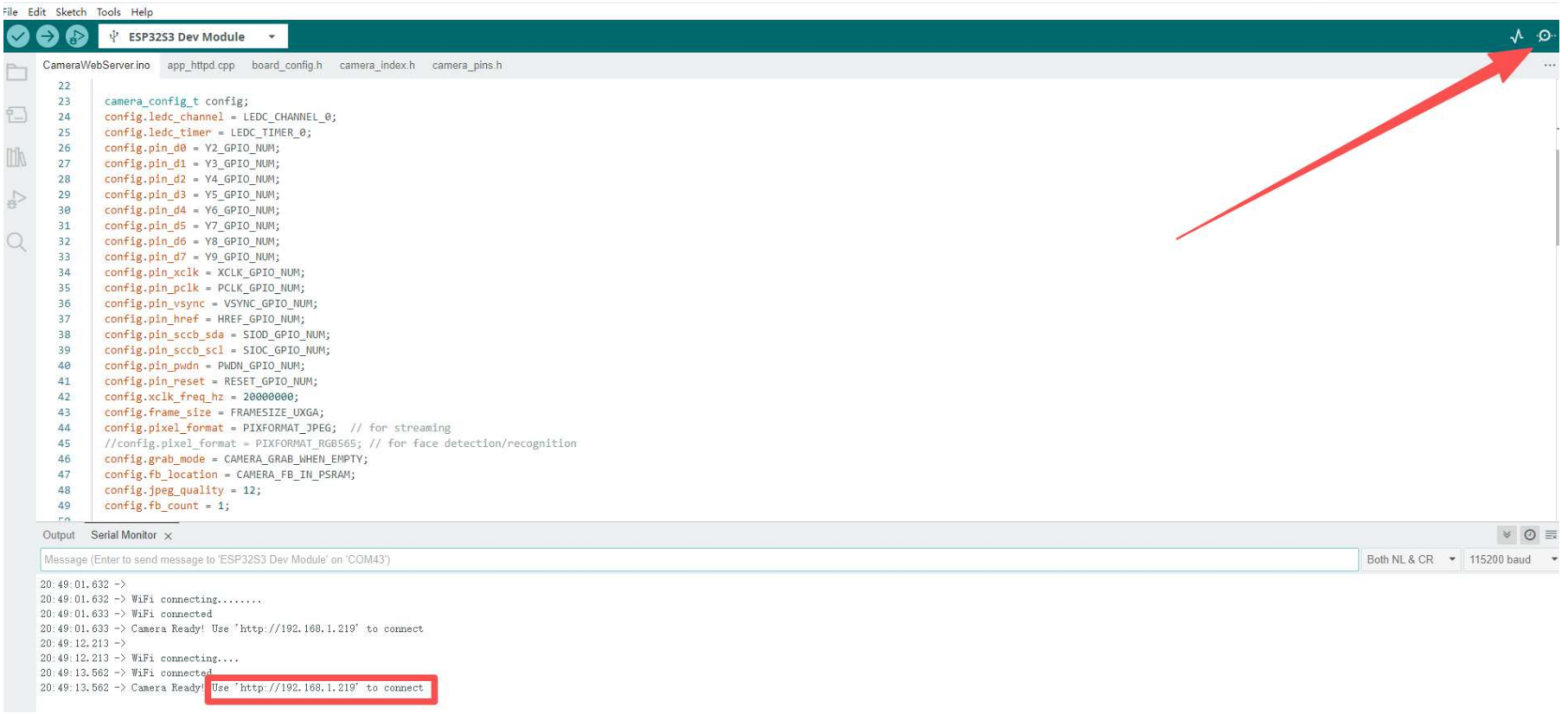

IR_Receiver | Arduino IDE 2.3.2
File Edit Sketch Tools Help
Auto Format Ctrl+T
Archive Sketch
Manage Libraries... Ctrl+Shift+I
Serial Monitor Ctrl+Shift+M
Serial Plotter
Firmware Updater
Upload SSL Root Certificates
Board: "ESP32S3 Dev Module"
Port: "COM3"
Get Board Info
USB CDC On Boot: "Enabled"
CPU Frequency: "240MHz (WiFi)"
Core Debug Level: "None"
USB DFU On Boot: "Disabled"
Erase All Flash Before Sketch Upload: "Disabled"
Events Run On: "Core 1"
Flash Mode: "QIO 80MHz"
Flash Size: "8MB (64Mb)"
JTAG Adapter: "Disabled"
Arduino Runs On: "Core 1"
USB Firmware MSC On Boot: "Disabled"
Partition Scheme: "8M with spiffs (3MB APP/1.5MB SPIFFS)"
PSRAM: "OPI PSRAM"
Upload Mode: "UART0 / Hardware CDC"
Upload Speed: "921600"
USB Mode: "Hardware CDC and JTAG"
Zigbee Mode: "Disabled"
Programmer
Burn Bootloader

Configure Wi-Fi: In the code, replace the values of the SSID and PASSWORD variables with your own 2.4GHz Wi-Fi credentials.




Enable Serial Communication:

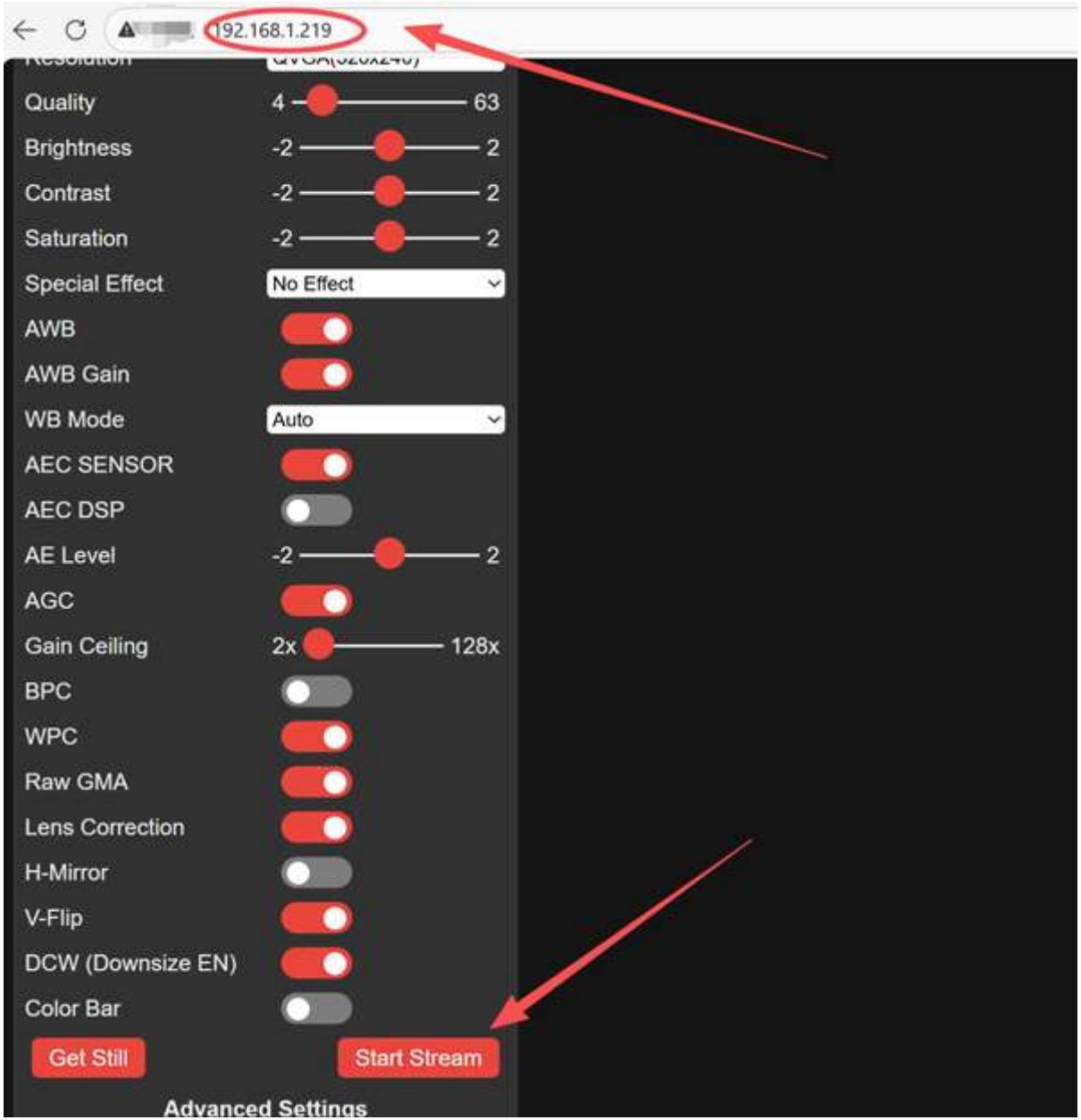
After uploading the program, open the Serial Monitor to view device logs and status.

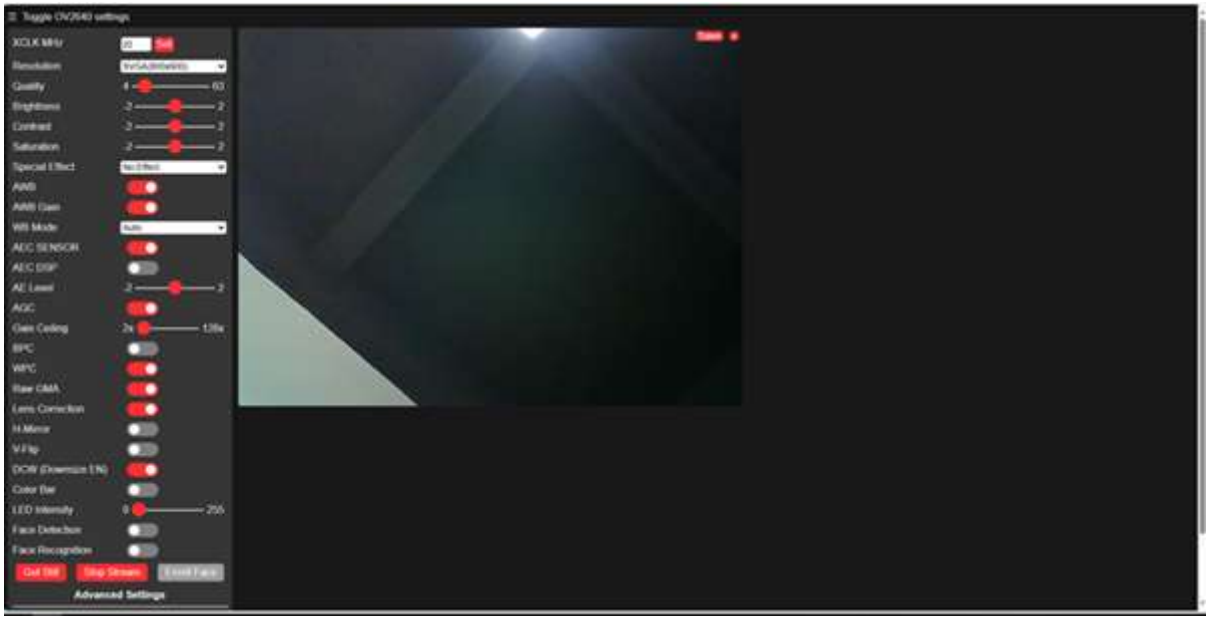



[Download the Arduino example]

## Result

Open the serial monitor, open the web page, Enter the address information displayed by the serial port in the browser.


192.168.1.219
Quality 4 63
Brightness -2 2
Contrast -2 2
Saturation -2 2
Special Effect No Effect
AWB
AWB Gain
WB Mode Auto
AEC SENSOR
AEC DSP
AE Level -2 2
AGC
Gain Ceiling 2x 128x
BPC
WPC
Raw GMA
Lens Correction
H-Mirror
V-Flip
DCW (Downsize EN)
Color Bar
Get Still
Start Stream
Advanced Settings



# MicroPython Programming Guide for ESP32-S3-CAM

This guide provides a quick start for using MicroPython on the **emakefun ESP32-S3-CAM**. Our board is fully compatible with standard ESP32-S3-CAM firmware and existing online tutorials.

## Resources & Downloads

- Download MicroPython Firmware

## Recommended IDEs

We recommend the following tools for a seamless experience:

- **Thonny IDE:** Best for beginners. It has built-in support for ESP32S3.
- **VS Code (with Pymakr extension):** Best for professional developers.

# Camera API (MicroPython)

This document describes the API and configuration for the ESP32-S3-CAM camera module.

## Core Functions

### `camera.init(*)`

Initializes the camera module, The camera must be initialized before it can be used.

- **Key Parameters**
  - format: Image data format

| Format | Description |
|---|---|
| `camera.JPEG` | JPEG format (default) |
| `camera.YUV422` | YUV422 |
| `camera.RGB565` | RGB565 |
| `camera.GRAYSCALE` | Grayscale |

  - framesize: Resolution (Frame size)

| Resolution (Frame size) | Description |
|---|---|
| `camera.FRAME_96X96` | 96×96 |
| `camera.FRAME_QQVGA` | 160×120 |
| `camera.FRAME_QCIF` | 176×144 |

| Resolution (Frame size) | Description |
| --- | --- |
| camera.FRAME_HQVGA | 240×160 |
| camera.FRAME_240X240 | 240×240 |
| camera.FRAME_QVGA | 320×240 |
| camera.FRAME_CIF | 352×288 |
| camera.FRAME_HVGA | 480×320 |
| camera.FRAME_VGA | 640×480（default） |
| camera.FRAME_SVGA | 800×600 |
| camera.FRAME_XGA | 1024×768 |
| camera.FRAME_HD | 1280×720 |
| camera.FRAME_SXGA | 1280×1024 |
| camera.FRAME_UXGA | 1900×1200 |
| camera.FRAME_FHD | 1920×1080 |
| camera.FRAME_P_HD | 2560×1440 |
| camera.FRAME_P_3MP | 2048×1536 |
| camera.FRAME_QXGA | 2048×1536 |
| camera.FRAME_QHD | 2560×1440 |
| camera.FRAME_WQXGA | 2560×1600 |
| camera.FRAME_P_FHD | 2560×1600 |

| Resolution (Frame size) | Description |
| --- | --- |
| `camera.FRAME_QSXGA` | 2560×2048 |

- quality: JPEG quality. Default: 12, Range: 0~63
- d0: D0 pin. Default: 11 (ESP32S3 CAM pin value)
- d1: D1 pin. Default: 9 (ESP32S3 CAM pin value)
- d2: D2 pin. Default: 8 (ESP32S3 CAM pin value)
- d3: D3 pin. Default: 10 (ESP32S3 CAM pin value)
- d4: D4 pin. Default: 12 (ESP32S3 CAM pin value)
- d5: D5 pin. Default: 18 (ESP32S3 CAM pin value)
- d6: D6 pin. Default: 17 (ESP32S3 CAM pin value)
- d7: D7 pin. Default: 16 (ESP32S3 CAM pin value)
- vsync: VSYNC pin. Default: 6 (ESP32S3 CAM pin value)
- href: HREF pin. Default: 7 (ESP32S3 CAM pin value)
- pclk: PCLK pin. Default: 13 (ESP32S3 CAM pin value)
- pwdn: POWER pin. Default: -1 (not used)
- reset: RESET pin. Default: -1 (not used)
- xclk: XCLK pin. Default: 15 (ESP32S3 CAM pin value)
- sda: SDA pin. Default: 4 (ESP32S3 CAM pin value)

    - scl: SCL pin. Default: 5 (ESP32S3 CAM pin value)
    - xclk_freq: XCLK signal frequency. Default: camera.XCLK_10MHz, Options: camera.XCLK_20MHz
    - fb_size: Number of frame buffers. Default: 1
    - fb_location: Frame buffer location. Default: camera.DRAM, Options: camera.PSRAM
- **Parameter:** `type` specifies the camera sensor model (e.g., `camera.OV2640`).
- **Return:** Returns `True` if initialization is successful, `False` otherwise.

**`camera.deinit()`**